

Abuse of Apple Air Tags in Cyberstalking

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Abuse of Apple Air Tags in Cyberstalking, Auto Theft & IoT-Enabled Crime - A Technical & Behavioral Security Analysis

Executive Summary

Consumer-grade tracking devices like Apple Air Tags present dual-use risks helpful for locating lost items but increasingly misused for stalking, auto theft, and digital surveillance. This whitepaper analyzes the technical vulnerabilities, criminal methods, and systemic weaknesses that enable these abuses.

Introduction

This research explores how Air Tags function, their privacy model, and their misuse in real-world stalking and vehicular targeting incidents.

Technical Overview of AirTags

AirTags operate via Bluetooth LE, the U1 Ultra-Wideband chip, and Apple's crowdsourced Find My network. Their anonymous, encrypted nature, while privacy preserving, also creates blind spots for victims.

Practical Uses of Air Tags

Air Tags supports legitimate tracking scenarios such as asset recovery, pet tracking, and accessibility support for the visually impaired.

Cyberstalking Landscape

Stalking has evolved from physical pursuit to digitally assisted tracking using Bluetooth, social media, GPS, and data-broker information.

Case Studies of Abuse

Documented incidents include domestic violence victims discovering Air Tags hidden in vehicles, and auto theft operations placing tags on high-end cars for later retrieval.

Technical Vulnerabilities

Security weaknesses include firmware modification, silent mode attacks via speaker removal, BLE-based cloning, and exploitation of the Find My network.

Systemic Weaknesses

Apple's anti-stalking alerts remain inconsistent across platforms. Android users receive delayed or no Notifications.

Mitigation Recommendations

Manufacturers must enforce cross-platform alerts, stronger firmware integrity, and non-disableable safety tones. Users should practice BLE scanning and report suspicious tracking.

Conclusion

AirTags epitomize dual-use IoT technology. Stronger design controls, coordinated regulation, and public awareness are essential to reduce digital stalking risks.